

Statistical Review and A/B Testing for New York City TLC Project

Executive summary report
Commission Prepared by **Automatidata**

Overview

This project analyzes New York City TLC taxi trip data to find ways to increase revenue for drivers. Specifically, we conducted a statistical A/B test to analyze the relationship between payment type and fare amount.

Problem

Taxi drivers want to maximize their daily earnings, but it is unclear if the way a customer pays affects how much they spend. We need to know if credit card users pay higher fare amounts than cash users, and if this difference is real or just random.

Solution

The difference is real and not a random fluke. Customers who pay with credit cards do, on average, pay higher fare amounts than those who pay with cash.

Details

We analyzed clean trip records specifically focusing on `fare_amount` and `payment_type`.

We used a 5% significance level (0.05) for the two-sample t-test.

The t-statistic was approximately 6.87, and the p-value was extremely close to zero, leading us to strongly reject the null hypothesis.

Next Steps

Make the card payment screen fast and easy, with preset tipping buttons to encourage larger card transactions.

Share these findings with drivers to show them that accepting cards leads to higher fares, which easily covers any small card processing fees.

